Does Population Aging Affect CO_2 Emissions? A Global Study Based on the EKC Framework

Introduction

Carbon dioxide (CO2) has existed since the Earth's formation. It is naturally produced through processes such as plant photosynthesis and volcanic eruptions (Rae et al., 2021). However, with the emergence of human and the rise of human activity, especially industrialization, carbon emissions have significantly increased (Gasser et al., 2015). Economic growth and industrial expansion have led to increased energy use and higher CO_2 emissions, making it a key driver of environmental degradation (Mitic & Fedajev et al., 2022; Raupach et al., 2007; Alvarez & Montanes, 2023). Yet, as income levels rise, many developed countries implement strategies to reduce emissions. Balancing economic growth with environmental sustainability has become one of the major challenges in climate change research (Stern, 2007).

To explore this complex relationship (economic growth and CO_2 emissions), researchers have proposed models such as the Environmental Kuznets Curve (EKC) and the IPAT model. Both frameworks explore how economic activity influences environmental outcomes.

The Environmental Kuznets Curve (EKC) is one of the most widely used models for analyzing the relationship between economic growth and environmental quality (Zou et al., 2022). Originally introduced by Simon Kuznets (1955), the model suggests that environmental degradation rises during the early stages of economic development but begins to decline when income reaches a certain level. This leads to an inverted U-shaped relationship between income and environmental impact, some recent studies (e.g., Jahanger et al., 2023) have introduced a cubic term for income, leading to an N-shaped curve where emissions increase, then decrease, and rise again at higher income levels.

Before the Environmental Kuznets Curve (EKC), a different framework was proposed by Ehrlich and Holdren (1971) and later expanded by Richard T. Carson (2011), the IPAT model. This model which is a simplified version of the EKC states that environmental impact (I) is the product of population (P), affluence (A), and technology (T), highlighting that socio-demographic factors, such as population and urbanization, are also important drivers of environmental quality. However, one factor that has received relatively little attention is population aging (Kim, Choi, & Lee, 2020). Within this framework, we explore whether population aging contributes to a reduction in CO_2 emissions or, conversely, leads to an increase.

Kim, Lim, and Jo (2020) examined the relationship between population aging and CO_2 emissions in South Korea, finding that an increase in the elderly population was associated with lower emissions. However, it remains unclear whether this relationship holds globally. We have extended their work by analyzing panel data from 165 countries between 2010 and

2020 to examine the CO2 emissions from the transport and residential sectors and their association with age structures in both high and low-income countries.

The Cartografía Carbono 2020 study (Planet App) highlights that carbon footprints rise from age 15 across all categories (housing, transport, food, and consumption), peaking between ages 45 and 64. This reflects increased economic activity and consumption during the working years. After age 65, emissions tend to decline, especially from transportation, due to reduced travel and car usage. However, residential emissions may increase, as older individuals spend more time indoors, leading to higher electricity and heating usage.

Residential CO_2 emissions also vary significantly by both income level and age group. High-income countries and working-age populations (15–64) tend to have higher emissions than low-income countries and older individuals respectively. Several empirical studies support these findings. For example, by having more production activity, the working-age population boost energy consumption and hence increases CO2 emissions (Stéphane Mbiankeu Nguea 2023). Hamza and Gilroy (2010) found that younger individuals tend to have a greater preference for travel and durable goods, behaviors strongly linked to increased pollution. Similarly, Long and Feng et al. (2023) concluded that the working-age group has the highest average carbon footprint, while the elderly population (65+) has the lowest carbon footprints.

Furthermore, when considering income levels, we observe that high-income countries emit significantly more CO2 per capita than low-income countries in both the transport and residential sectors compared to low-income nations. To better illustrate this, we use histogram plot (Figure 1) to compare residential and transport-related CO2 emissions across different age groups and income levels in high, middle, and low-income countries.

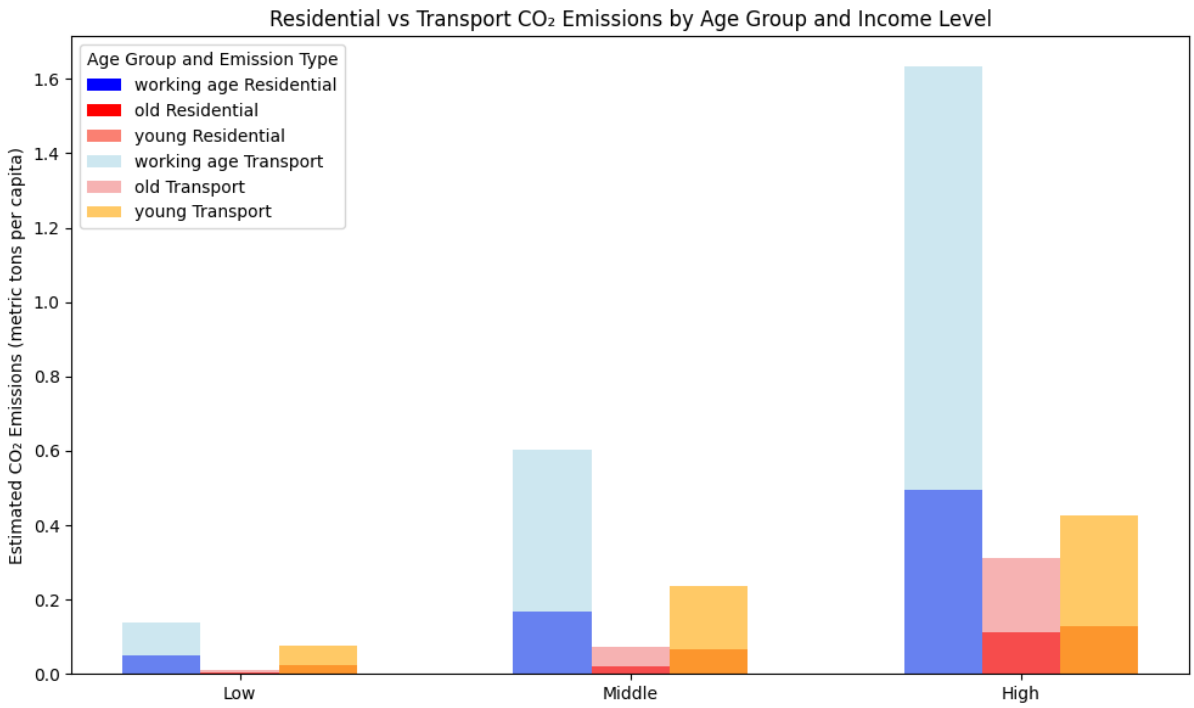

Figure 1: Residential and Transport CO_2 Emissions by Age Group and Income Level

Some recent studies have confirmed these findings. For example, Apeti et al. (2025) highlight that global CO_2 emissions are not distributed equally among the world's population. According to the United Nations Development Programme (UNDP, 2022), the poorest 50% of people are responsible for only 12% of global emissions, while the richest 10% account for as much as 48%. This highlights the need to consider both income and demographic differences when analyzing environmental impacts.

in this study we have examined the long-run effects of GDP and population aging on CO_2 emissions by Environmental Kuznets Curve (EKC). We have investigated that for both high-income and low-income countries how CO_2 emissions behave across different stages of economic development. In addition, we have examined that how changes in population such as aging, affect CO_2 emissions.

Our findings are consistent with Marselina and Prasetyo (2023), who show that more prosperous a country, the less the environmental degradation, while in low-income countries, carbon emissions increase when economic growth increases. However, some studies suggest that this relationship may not always follow a simple inverted U-shape. For example, Bruyn et al. (1998) propose an N-shaped EKC, where emissions initially rise with income, then fall, but may rise again at very high-income levels due to increased consumption or technological obsolescence.

Given the time-series nature of our dataset, we start by testing for unit roots to check whether the variables are stationary. To ensure the robustness of our results, we apply first-generation panel unit root tests, such as the CIPS, LLC, IPS, and Maddala-Wu tests (Levin et al., 2002; Im, Pesaran , 2007; Maddala & Wu, 1999). These tests assume cross-sectional independence, meaning countries are treated as unaffected by each other. However, this assumption is often unrealistic in macroeconomic contexts, where countries are interconnected through trade, finance, and environmental impacts.

To address these limitations, we apply second-generation panel cointegration techniques that account for both cross-sectional dependence and heterogeneity. Specifically, we employ the Pedroni test (1999), which allows for heterogeneous cointegration relationships across countries, and the Kao test (1999), which assumes a homogeneous long-run relationship based on a residual-based extension of the Engle–Granger approach. These methods help us examine the long-run relationship between CO_2 emissions, economic growth, and population aging more accurately, reflecting the interdependence of countries in the global economy.

Once cointegration isconfirmed, we use the Fully Modified Ordinary Least Squares (FMOLS) method (Pedroni, 1996, 2000) to estimate long-run relationships. FMOLS is suited for panel data analysis with time-series properties, as it corrects for issues like autocorrelation and endogeneity, providing more reliable estimates of the long-run effects of economic and demographic factors on CO_2 emissions.

This study aims to examine these long-run relationships at the global level, as well as across high-income and low-income countries, to better understand how economic development and population structure influence total and sector-specific emissions, residential and transport, during the period 2010 to 2020.

2. DATA and Methodology

In this study, data for all variables were collected from the 'World Bank' and 'Our World in Data' sources, covering 165 countries from 2010 to 2020. The dataset includes environmental, economic, and demographic indicators relevant to the Environmental Kuznets Curve (EKC) and IPAT frameworks.

CO_2 emissions per capita is considered as the dependent variable, representing each country's contribution to global emissions. The key explanatory variables include GDP per capita (as an economic indicator), population age structure (comprising the working-age population aged 15–64, young population aged 0–14, and elderly population aged 65+). Additionally, we incorporate urban population and total population as demographic variables.

Given the unavailability of oil price data for all countries (which was used in Kim, Lim, and Jo's 2020 study), we use renewable energy consumption (% of total energy use) and primary energy consumption per capita as proxy variables to analyse energy trends. These proxies help capture the energy mix and overall consumption patterns that influence CO_2 emissions.

Manufacturing value added (% of GDP) represents the contribution of the manufacturing sector to a country's GDP, while services value added (% of GDP) reflects the proportion of GDP generated by the services sector, including sectors like finance, education, healthcare, and tourism.

we also consider CO_2 emissions from transport. For CO_2 emissions from the residential sector, we use per capita carbon dioxide emissions from buildings as a proxy. This variable captures the emissions associated with energy consumption in residential buildings, which is a key contributor to household carbon footprints.

Before analysis, the data were cleaned and checked for consistency. All variables were converted into their logarithmic form to reduce skewness and to interpret the results in correct way. The data is balanced so that all countries have data available for the period from 2010 to 2020.

3. Empirical Analysis

In this section, we set up the empirical model using a balanced panel dataset covering 165 countries over the period 2010 to 2020. To explore the nonlinear relationship between economic growth and CO_2 emissions, we include a cubic term of GDP per capita and a set of demographic variables to estimate an Environmental Kuznets Curve (EKC) model. This extended model allows us to test for both the traditional inverted U-shaped EKC and a potential N-shaped pattern, while also assessing the long-run effects of population aging and other demographic factors on emissions. in other words, the nonlinear relationship between income and emissions can be estimated in two polynomial regression models:

(**Quadratic EKC model):** a second-degree model to test the classic inverted U-shaped Environmental Kuznets Curve (EKC),

$\ln(CO2)_{it}=\alpha+\beta_1 \ln(GDP_{it})+\beta_2 \ln(GDP_{it})^2+\varepsilon_{it}$

(Cubic (N-shaped) EKC model): a third-degree model to explore the possibility of an N-shaped relationship,

$\ln(CO2)_{it}=\alpha+\beta_1 \ln(GDP_{it})+\beta_2 \ln(GDP_{it})^2+\beta_3 \ln(GDP_{it})^3+\varepsilon_{it}$

where ε_{it} is the error term for country *i* in year *t*.

Moreover, As previously discussed, we use the Fully Modified Ordinary Least Squares (FMOLS) method to estimate the long-run relationship between CO_2 emissions and its main drivers. Unlike standard OLS, FMOLS is designed for panel data with non-stationary variables, as it addresses issues of autocorrelation and endogeneity, thereby it provides more reliable and consistent estimates.

3.1. base model

The baseline model used in this study includes total population (P), income per capita (GDP), and a range of exogenous variables. The model is specified in logarithmic form as follows:

$\log(CO2_{it})=\alpha_i+\beta_1\log(GDP_{it})+\beta_2\log(GDP_{it})^2+\beta_3\log(GDP_{it})^3+\beta_4\log(Energy_{it})+$

$\beta_5\log(Renew_{it})+\beta_6\log P_{it}+\beta_7\log(MAF_{it})+\beta_8\log(SVC_{it})+\beta_9\log(Young_{it})+ \beta_{10}\log(old_{it})+u_{it}$

Where:

- α_I represents unobserved country-specific effects.
- $Energy_{it}$ denotes primary energy consumption per capita.
- $Renew_{it}$ is the share of renewable energy consumption.
- MAF_{it} is the ratio of manufacturing value-added.
- SVC_{it} is the ratio of services value-added.
- $Young_{it}$ refers to the proportion of the population aged 0–14.
- old_{it} denotes the share of the population aged 65 and above.
- u_{it} is the error term for country i in year t.

The model includes a third-order polynomial of income to capture the nonlinearities between income and emissions. It also incorporates the Youth Dependency Ratio and the Old-Age Dependency Ratio as demographic variables. The Youth Dependency Ratio is defined as the ratio of the population aged 0–14 to the working-age population (15–64), while the Old-Age Dependency Ratio represents the share of those aged 65 and above relative to the working-age group. Table 1 presents descriptive statistics for the log-transformed variables used in the analysis.

(TABLE 1 Descriptive statistics)

	MEAN	STD	SKEWNESS	KURTOSIS	COUNT
LN_CO2	0.655588	1.506451	-0.429914	-0.573371	1791.0
LN_GDP	8.646631	1.419496	0.040226	-0.806392	1791.0
LN_ENERGY	9.126022	1.905334	-2.451578	14.728802	1791.0
LN_RENEW	2.733482	1.919876	-2.069871	4.659089	1791.0
LN_P	15.770307	1.983467	-0.494550	0.877514	1791.0
LN_MAF	2.345987	0.565283	-1.048931	1.566132	1791.0
LN_SVC	3.986314	0.215636	-0.918758	1.760218	1791.0
LN_OLD	1.845177	0.755953	0.120209	-1.156619	1791.0
LN_YOUNG	3.272566	0.400434	-0.141712	-1.322283	1791.0
LNYOUTHDEP	-0.841283	0.481158	0.104312	-1.277395	1791.0
LNOLDDEP	-2.180748	0.637661	0.243346	-0.921614	1791.0

We use the logarithms of the variables to reduce skewness and stabilize variance. Even after this transformation, the variables still show different patterns. For example, the log of CO_2 emissions per capita (ln_CO2) has a slight negative skew (−0.43), meaning that more countries have emissions levels higher than the average. Energy use and renewable energy remain highly skewed, showing that there are still big differences between countries. In contrast, the economic and demographic variables are more symmetrically distributed, which supports the use of log-linear models in the analysis. Overall, these results show the importance of taking country differences into account when studying what affects CO_2 emissions.

3.2. Panel Unit Root Tests

Before estimating the main model, we first test the stationarity of the variables, as using non-stationary data in long-run regressions such as FMOLS may lead to misleading results. We begin by applying first-generation panel unit root tests, Levin-Lin-Chu (LLC), Im-Pesaran-Shin (IPS), and Maddala-Wu (MW), which are widely used in panel data studies. However, these tests assume cross-sectional independence, meaning that countries are not influenced by each other. This assumption is often unrealistic in macroeconomic data, especially during globally impactful periods like 2010–2020, which included international economic shocks and environmental policy changes. To address this issue, we apply the CIPS (Cross-sectionally Augmented IPS) test proposed by Pesaran (2007), which accounts for interdependence among countries by incorporating average cross-sectional effects.

The CIPS test statistic is calculated by averaging the individual t-statistics across all N cross-sectional units (countries):

$$CIPS(N,T)= N^{-1} \sum t(N,T)$$

Where:

- N = number of cross-sectional units (in our case, 165 countries)
- T = number of time periods (here, 11 years, from 2010 to 2020)
- t = the ADF t-statistic

The null hypothesis of the CIPS test is that the variable contains a unit root, meaning it is non-stationary. if the CIPS test statistic is more negative than the critical value at a chosen significance level, the null is rejected, indicating that the variable is stationary.

Pesaran (2007) proposed a table which contains the critical values from for different values of N, T, and significance levels. The relevant values for our dataset (with N = 165 and T = 11) are similar to the row for N = 200, T = 10–20, so critical values around –2.23 (1%), –2.12 (5%), and –2.05 (10%) can be used as reference thresholds.

Table 2: ADF Unit Root Test Statistics

variables	NO Trend	trend
ln_CO2	-1.393	-2.266
ln_GDP	-2.113	-2.694
ln_energy	-2.039	-2.289
ln_renew	-1.459	-2.378
ln_P	-1.905	-1.329
ln_SVC	-4.652	-2.285
ln_MAF	-1.630	-2.543
ln_old	-2.477	-2.836
ln_young	-1.223	-2.494
lnYouthDep	-1.040	-2.551
lnOldDep	-2.266	-2.899

table 2 presents the CIPS (Cross-sectionally Augmented IPS) test statistics for each variable, both with and without a linear trend.

Based on the critical values that we found, we compare each variable's CIPS test statistic with the thresholds. If a variable's statistic is more negative than, for example, –2.12 (5% level), we can reject the null and conclude that the variable is stationary. The results indicates that some variables, like ln_old and ln_services, are stationary based on the CIPS test, while others are not. The test results indicate mixed stationarity and non-stationary, we treat all variables as non-stationary and continue with cointegration analysis using the Fully Modified Ordinary Least Squares (FMOLS) method, which allows us to estimate valid long-run relationships even when some variables have trends over time. This approach ensures a robust examination of how GDP, aging, and other factors influence CO_2 emissions over the long term in different countries.

3.3. Panel Cointegration Test

After confirming that many variables in our dataset are non-stationary, the next step is to check whether the variables move together in a stable way over the long term, even though they change over time. This concept is known as cointegration. To examine this, we apply

panel cointegration tests developed by Pedroni and Kao, which are useful for macro panel data that involve multiple countries over time.

Pedroni's test is for heterogeneous dynamics, meaning that each country in the panel can have its own short-run behavior, while still testing for the existence of a common long-run relationship among the variables. Pedroni also introduces both within-dimension (panel) and between-dimension (group) test statistics, such as the panel ADF and group ADF tests, which allow for heterogeneous dynamics across cross-sectional units. This means that each country can have its own adjustment process and short-term behavior, making the tests well-suited for macro panel data where countries differ in economic and demographic conditions.

On the other hand, Kao test assumes homogeneous cointegrating vector across all panel units, meaning that countries are expected to share the same long-run relationship. in fact, this test is a complementary robustness check to the Pedroni test. Its inclusion helps validate the existence of long-run equilibrium relationships among variables such as CO_2 emissions, GDP per capita, and age structure, under the assumption of homogeneous dynamics.

Table 3. Kao cointegration test results

	MDF	DF	ADF	UMDF	UDF
statistics	-1.98	-1.99	-2.68	-2.0	-1.99

All five Kao test statistics are more negative than −1.95, which means they are statistically significant at the 5% level or better. One of them (ADF = −2.68) is even significant at the 1% level. this provides strong evidence that the variables like CO_2 emissions, GDP per capita, age structure, are cointegrated, which means they move together over the long run.

We have the same results from Pedroni Cointegration Test, when we include a time trend, all test statistics reject the null hypothesis, meaning, there is strong evidence that the variables are cointegrated. Table 4 presents the Pedroni-style ADF cointegration test statistics. The Panel ADF test assumes homogeneous dynamics across countries, while the Group ADF test allows for heterogeneity, meaning each country can follow its own short-run adjustments. Trend specification provides stronger support for cointegration, especially for the Group ADF test. Since the Group ADF statistic with trend (–2.86) is lower than 5% critical value (around –2.12), it indicates a strong long-run relationship among the variables.

Table 4. Pedroni cointegration test results

	No Trend	Trend
Panel ADF	-1.393	-2.266
Group ADF	-1.993	-2.866

To ensure robustness, we apply both the Pedroni ADF-based tests and the Kao residual-based cointegration test. The Group ADF (with trend) is significant at the 5% level, and Kao's test also supports cointegration at the 10% level. Together, these results confirm the existence of a long-run equilibrium relationship between CO_2 emissions and the key economic and demographic variables across countries.

3.4. Panel Fully Modified OLS (FMOLS) Estimation

In panel data analysis, one major concern is the non-stationarity of time-series variables. A common way to handle this issue is by converting the variables into first differences (using the change from one year to the next), which makes the series stationary. However, this method captures only short-run effects, it is less suitable for long-run relationships.

Since our study focuses on long-term relationships, such as the Environmental Kuznets Curve (EKC) and the impact of population aging on CO_2 emissions, it is important to estimate long-run effects, not just short-run fluctuations.

To address this, we apply the Fully Modified Ordinary Least Squares (FMOLS) which is designed specifically for cointegrated panel data and provides unbiased and consistent estimates of long-run relationships, even when the variables are non-stationary.

To explore the long-run relationship between CO_2 emissions, economic development, and demographic factors, we estimate six different FMOLS models, each consists of different variable combinations to test the robustness of our findings. Models (1) and (2) include a cubic term for GDP per capita (GDP^3) to test the N-shaped EKC hypothesis, allowing for non-linear relationships between income and emissions. Both models also include population, energy consumption, and the share of services in GDP. Model (1) uses the share of young and elderly populations, while Model (2) uses dependency ratios (youth-to-working-age and elderly-to-working-age) to compare the effects.

Models (3) and (4), EKC with quadratic income, Same as Models (1) and (2), but remove the cubic income term. This allows for a simpler U-shaped or inverted U-shaped EKC (not N-shaped). Models (5) and (6) are for robustness checks. Model (5) excludes Services variable and Model (6) removes both manufacturing and Services allowing us to assess the model without industrial structure variables.

To account for heterogeneity across economic conditions, we estimate the FMOLS models for three samples:

1. The global dataset (165 countries),
2. A subset of high-income countries (GDP per capita ≥ $30,000),
3. A subset of low-income countries (GDP per capita < $10,000).

3.4.1. FMOLS results: Global Dataset

We begin our empirical analysis by estimating the Fully Modified Ordinary Least Squares (FMOLS) models using the global dataset. This approach allows us to examine the long-run relationship between per capita CO_2 emissions and key factors such as economic growth, energy consumption, demographic structure, and industrial composition. Table 5 reports the results from six model specifications that we explained them before, each designed to test the robustness of these relationships across different combinations of explanatory variables.

Table5: Panel fully modified ordinary least squares (FMOLS) estimation results all countries:

VAR	MODEL (1)	MODEL (2)	MODEL (3)	MODEL (4)	MODEL (5)	MODEL (6)
LN_GDP	5.88 ***	5.68 ***	2.09 ***	2.07 ***	6.06 ***	5.72 ***
LN_GDP2	-0.54 ***	-0.52 ***	-0.09 ***	-0.09 ***	-0.57 ***	-0.53 ***
LN_GDP3	0.02 ***	0.02 ***	NAN	NAN	0.02 ***	0.02 ***
LN_ENERGY	0.17 ***	0.17 ***	0.17 ***	0.17 ***	0.17 ***	0.17 ***
LN_RENEW	-0.24 ***	-0.23 ***	-0.23 ***	-0.22 ***	-0.24 ***	-0.23 ***
LN_YOUNG	-0.37 ***	NAN	-0.37 ***	NAN	-0.40 ***	NAN
LN_OLD	0.22 ***	NAN	0.19 ***	NAN	0.18 ***	NAN
LN_P	0.02 ***	0.02 ***	0.02 ***	0.02 ***	0.02 ***	0.03 ***
LN_SVC	-0.34 ***	-0.33 ***	-0.36 ***	-0.35 ***	NAN	NAN
LN_MAF	0.09 ***	0.08 ***	0.08 ***	0.08 ***	0.09 ***	NAN
LNYOUTHDEP	NAN	-0.42 ***	NAN	-0.43 ***	NAN	-0.45 ***
LNOLDDEP	NAN	0.21 ***	NAN	0.17 ***	NAN	0.17 ***
ADJ-R2	0.92	0.92	0.92	0.92	0.92	0.92
OBS	1791	1791	1791	1791	1791	1791
TP (USD)	$221	$232	$90,316	$91,452	$206	$225

the results from all models support the Environmental Kuznets Curve (EKC) hypothesis. The coefficients for ln_GDP and ln_GDP^2 are positive and negative, respectively, and statistically significant at the 1% level. This implies an inverted U-shaped relationship between income and CO_2 emissions. This suggests that emissions initially rise with economic growth but begin to decline after reaching a certain income level. Additionally, the positive and significant cubic GDP term (ln_GDP^3) in models (1,2,5,6) implies an N-shaped EKC, which means at very high-income levels, emissions may rise again.

Energy consumption (ln_energy) is positively associated with CO_2 emissions, while renewable energy consumption (ln_renew) is negatively associated, both significant at the 1% level. These findings align with expectations and highlight the critical role of energy policy in environmental sustainability (Qudrat-Ullah &Nevo, 2021)

Demographic variables also show significant effects across the models. Young population (ln_young in Models 1, 3, and 5, lnYouthDep in Models 2, 4, and 6) is negatively associated with CO_2 emissions. This suggests that youth population tend to emit less, because younger individuals (aged 0–14) engage in less economic activity and consume fewer resources than adults,(*Cartografía,Carbono,2020).*
In contrast, the elderly population (ln_old and lnOldDep) shows a positive relationship with emissions across all models. This may reflect increased residential energy use or the services demand among older populations. These findings indicate the importance of population age structure in influencing emission levels, highlighting that both young and aging demographics play distinct roles in environmental outcomes.

All models demonstrate strong explanatory power, with adjusted R^2 values of 0.92, indicating that the selected variables explain 92% of the variation in per capita CO_2 emissions.

Figures 2 and 3 illustrate the Environmental Kuznets Curve (EKC) using global panel data. These plots are consistent with the FMOLS regression models in table 5.

In the quadratic models (Models 3 and 4), the estimated turning points are around $90,000–$91,000. Since most countries in the dataset have lower GDP per capita, this explains the strong positive relationship between GDP and emissions observed in Figure 2, with only a slight flattening at higher income levels. In contrast, the cubic models (Models 1, 2, 5, and 6) indicate much lower turning points, around $221–$344. This implies that most countries are already on the second segment of an N-shaped curve, for this reason the N-shape appears as a monotonic rise with a slight flattening in Figure 3 rather than a complete curve.

While the statistical models support both inverted U-shaped and N-shaped EKCs, the full patterns are not entirely visible due to the position of turning points. The inverted U-curve remains incomplete because of high thresholds, and the early decline of the N-curve is not fully captured. These results highlight that many developing and middle-income countries are still in the emission-increasing stage of the income-emissions relationship.

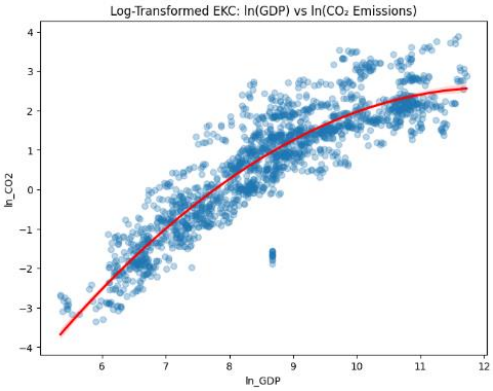

Figure 2

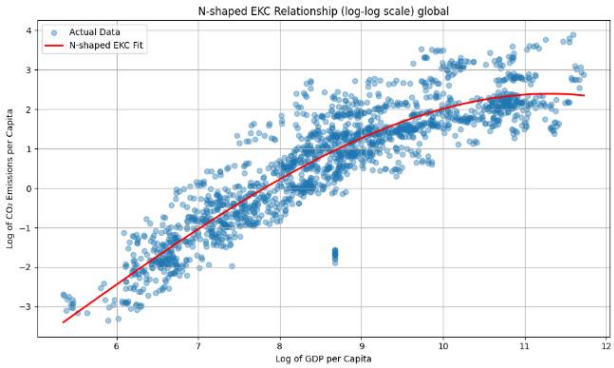

Figure 3

3.4.2. FMOLS results: High-Income Countries

To better understand how economic and demographic factors affect CO_2 emissions in more developed economies, we apply the FMOLS estimation to a subset of high-income countries (defined as those with GDP per capita above $30,000). This analysis helps reveal whether the Environmental Kuznets Curve (EKC) and the influence of population aging behave differently in wealthier settings compared to the global sample.

Table 6: Panel fully modified ordinary least squares (FMOLS) estimation results for high-income countries:

VAR	MODEL (1)	MODEL (2)	MODEL (3)	MODEL (4)	MODEL (5)	MODEL (6)
LN_GDP	810.60 ***	818.70 ***	-5.18	-5.24	799.84 ***	842.30 ***
LN_GDP2	-74.15 ***	-74.89 ***	0.25	0.25	-73.16 ***	-77.19 ***
LN_GDP3	2.26 ***	2.28 ***	NAN	NAN	2.23 ***	2.36 ***
LN_P	0.11 ***	0.11 ***	0.11 ***	0.11 ***	0.11 ***	0.11 ***
LN_ENERGY	0.40 ***	0.39 ***	0.41 ***	0.41 ***	0.39 ***	043 ***
LN_RENEW	-0.03	-0.03	-0.04 **	-0.05 **	-0.03	-0.03
LN_YOUNG	0.27 **	NAN	0.31 ***	NAN	0.26 **	NAN
LN_OLD	-0.28 ***	NAN	-0.24 ***	NAN	-0.32 ***	NAN
LN_SVC	-0.23	-0.27	-0.18	-0.22	NAN	NAN
LN_MAF	-0.13 **	-0.13 **	-0.17 ***	-0.17 ***	-0.12 **	NAN
LNYOUTHDEP	NAN	0.25 **	NAN	0.30 ***	NAN	0.33 ***
LNOLDDEP	NAN	-0.32 ***	NAN	-0.28 ***	NAN	-0.38***
ADJ-R2	0.75	0.75	0.71	0.71	0.75	0.74
OBS	280	280	280	280	280	280
TP (USD)	$236	$236	$30,598	$30,665	$237	$234

The FMOLS results for high-income countries support an N-shaped Environmental Kuznets Curve (EKC). In Models (1), (2), (5), and (6), the GDP coefficients follow the expected nonlinear pattern, ln(GDP) is positive and highly significant, the squared term is negative, and the cubic term is positive and statistically significant. This combination confirms the existence of an N-shaped relationship between income and CO_2 emissions, where emissions first increase with economic growth, then decline, and eventually rise again at higher income levels.

The estimated turning points for these models lie between $234 and $237. This indicates that the full N-shaped pattern is not only statistically significant but also observable in the data. As shown in Figure 4, the traditional EKC hypothesis will not hold in the longterm. Increasing income might once again lead to a positive relationship between economic growth and environmental degradation (de Bruyn et al. 1998). This implies that, even in wealthy countries, economic growth does not always lead to improved environmental conditions in the long run. The impact of technological obsolescence alters the typical inverted U-shaped relationship into an N-shaped one. This is because outdated technologies can lead to increased pollution (Fakher&Ahmed et al 2023).

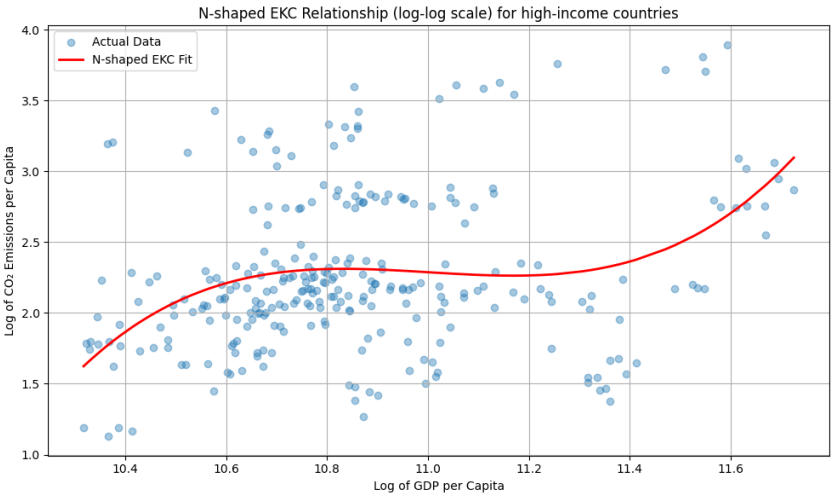

Figure 4: Environmental Kuznets Curve (EKC) for High-Income Countries

As we see in table 6 the other variables also contribute to explaining emissions. Energy use has a strong and positive effect, while renewable energy consumption is associated with lower emissions in all the models.

The demographic structure also demonstrates important effects. The young population (ages 0–14) shows a positive and statistically significant relationship with CO_2 emissions. On the other hand, the elderly population (65+) has a negative and significant effect on emissions.

This finding is consistent with Kim, Lim, and Jo (2020), who observed in South Korea that an increasing share of elderly individuals is associated with lower emissions. Older adults tend to have reduced mobility, spend more time at home, and live in smaller households. They also use fewer appliances and rely less on transportation, which altogether leads to lower energy consumption and, consequently, reduced CO_2 emissions.

Overall, the adjusted R^2 values between 0.71 and 0.75 show that the models explain a substantial portion of the variation in emissions. These findings suggest that even in high-income countries, where environmental policies are often stronger, emissions can increase at higher income levels.

3.4.3. FMOLS results: Low-Income Countries

Table 7 presents the FMOLS estimation results for low-income countries (GDP per capita < $10,000). This analysis allows us to explore how economic growth, energy use, and population structure affect CO_2 emissions in developing economies.

Table7: Panel fully modified ordinary least squares (FMOLS) estimation results for low-income countries:

VAR	MODEL (1)	MODEL (2)	MODEL (3)	MODEL (4)	MODEL (5)	MODEL (6)
LN_GDP	4.54	4.68	2.45 ***	2.35 ***	4.83 **	4.51
LN_GDP2	-0.40	-0.43	-0.12 ***	-0.11 ***	-0.44	-0.40
LN_GDP3	0.01	0.01	NAN	NAN	0.01	0.01
LN_P	0.02 **	0.02 **	0.02 ***	0.02 ***	0.02 ***	0.03 ***
LN_ENERGY	0.15 ***	0.15 ***	0.15 ***	0.15 ***	0.15 ***	0.15 ***
LN_RENEW	-0.28 ***	-0.28 ***	-0.28 ***	-0.28 ***	-0.29 ***	-0.27 ***
LN_YOUNG	-0.62 ***	NAN	-0.62 ***	NAN	-0.63 ***	NAN
LN_OLD	0.15 **	NAN	0.15 **	NAN	0.14 **	NAN
LN_SVC	-0.16 **	-0.16 **	-0.16 **	-0.17 **	NAN	NAN
LN_MAF	0.11 ***	0.10 ***	0.11 ***	0.10 ***	0.10 ***	NAN
LNYOUTHDEP	NAN	-0.59 ***	NAN	-0.58 ***	NAN	-0.62 ***
LNOLDDEP	NAN	0.14 **	NAN	0.14 **	NAN	0.12
ADJ-R2	0.89	0.89	0.89	0.89	0.89	0.89
OBS	1193	1193	1193	1193	1193	1193
TP (USD)	$288	$240	$30,206	$33,154	$240	$266

The FMOLS estimation results for low-income countries provide partial support for the Environmental Kuznets Curve (EKC) hypothesis. As we see the evidence for an N-shaped relationship is weak and largely theoretical. In Models (1), (2), (5), and (6), positive coefficients for the linear GDP term (ln_GDP), negative for the squared term (ln_GDP2), and positive for the cubic term (ln_GDP3). However, the coefficients for the squared and cubic terms are small and statistically insignificant.

What the plot exhibits is a rising trend for EKC, which is inconsistent with the expected N-shaped or traditional inverted U-shaped pattern (figure 5). The fitted curve displays a clear upward trend, with no visible turning point or second rise, despite the cubic specification. This trend might suggest that low-income countries are in an initial stage of economic development where growth is linked to increased emissions without a subsequent decline or secondary rise in emissions, could imply that these countries have not yet reached the economic threshold where the environmental improvements begin. This aligns with findings by Richmond and Kaufmann (2006) that in developing countries, the relationship remains consistently positive, showing no turning point.

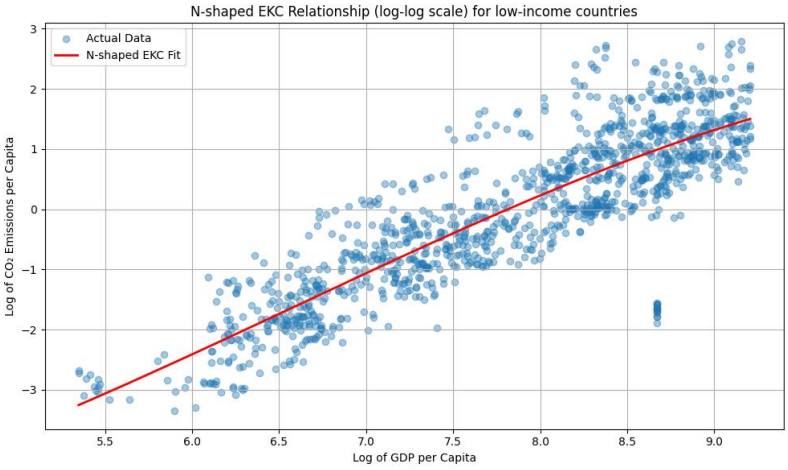

Figure 5: Environmental Kuznets Curve (EKC) for Low-Income Countries

This is reflected in the turning points, as we see in table (7), turning points in the cubic models appear at very low-income levels, between $240 and $288. These values are small so that most of these countries are still in the stage where CO_2 emissions rise as income increases. In contrast, the quadratic models show turning points between $30,000 and $33,000, which are much higher than the income levels of low-income countries. This implies that for nearly all countries in this group, CO_2 emissions are still increasing with income, and the downward of the EKC are not empirically observable and we do not observe the stage where emissions start to decline.

In terms of other variables in table 7 we observe that energy use is positively associated with emissions, as expected. Renewable energy consumption has negative effect, indicating its potential to reduce emissions in developing contries. Young population share (ln_young and lnYouthDep) is negatively related to emissions, while elderly population (ln_old and lnOldDep) has a small positive effect. All the models show high explanatory power, with adjusted R^2 values 0.89, indicating a strong fit to the data.

3.5.1. Time Series Fully Modified OLS

In earlier sections, our global FMOLS analysis showed that demographic structure, especially the shares of young and elderly populations, significantly affects long-run CO_2 emissions per capita. However, a limitation of the panel dataset is that it combines emissions from all

sectors, without separating specific sources like transportation or residential energy use. This might hide how different age groups cause emissions in different ways.

For instance, in countries where car use is common, younger people may produce more CO_2 from transportation because they travel more and consume more. On the other hand, older people may travel less and contribute less to transport emissions, but they might increase residential emissions since they spend more time at home, especially in places where heating or cooling is needed.

To address this, we extend our analysis by running separate time-series FMOLS regressions using two sector-specific dependent variables: CO_2 emissions from the transportation sector and from the residential sector. This allows us to explore whether the relationship between age structure and emissions differs across sectors.

For the global time series analysis, we estimated six FMOLS models to examine the long-run relationship between CO_2 emissions and key drivers across two sectors. Table 8 presents the results: Models RES (1), RES (2), and RES (3) focus on residential CO_2 emissions, while Models TR (4), TR (5), and TR (6) correspond to emissions from the transportation sector.

Table 8: Time series FMOLS estimation results for all the countries (global)

	RES (1)	RES (2)	RES (3)	TR (4)	TR (5)	TR (6)
CONST	-13.39 ***	-4.77 ***	-13.23 ***	-15.38 ***	-8.71 ***	15.37***
LN_GDP	0.04	0.62 ***	0.04	0.67 ***	1.02 ***	0.7***
LN_GDP2	0.0	-0.03 ***	0.0	-0.01 **	-0.03 ***	-0.02***
LN_P	0.11 ***	0.1 ***	0.1 ***	-0.03 ***	-0.03 ***	-0.02***
LN_ENERGY	0.18 ***	0.18 ***	0.18 ***	0.3 ***	0.3 ***	0.3***
LN_RENEW	-0.3 ***	-0.29 ***	-0.29 ***	-0.07 ***	-0.07 ***	-0.06***
LN_YOUNG	0.88 ***	NAN	0.9 ***	0.56 ***	NAN	0.53***
LN_OLD	1.34 ***	NAN	1.35 ***	0.25 ***	NAN	0.24***
LN_WORKING AGE	0.68 *	NAN	0.7 *	1.38 ***	NAN	1.38***
LNYOUTHDEP	NAN	0.18 **	NAN	NAN	-0.01	NAN
LNOLDDEP	NAN	1.26 ***	NAN	NAN	0.09 **	NAN
LN_SVC	NAN	NAN	-0.08	NAN	NAN	NAN
LN_MAF	NAN	NAN	NAN	NAN	NAN	0.07***
ADJ-R2	0.74	0.74	0.74	0.87	0.87	0.89
OBS	1791	1791	1791	1791	1791	1791

we find that age structure continues to play an important role in emissions. ln_young, ln_old, and ln_working age are positive and significant in both residential and transportation models. This suggests that larger population groups, regardless of age, are associated with higher emissions. However, in the residential sector, old population (*ln_old*) are strongly associated with CO_2 emissions, suggesting that older population contribute more to residential energy use, due to more time spent at home, higher appliance use as discussed in (Kim et al., 2020).

In contrast, for transportation emissions, younger populations (*ln_young*) and working age population show a positive strong relationship with emissions, indicating higher levels of mobility and travel activity among these groups which is in line with findings from [Liddle & Lung, 2010].

ln_P (total population) is positive in residential models, but negative in transportation models. This indicates that more people use public transport like buses and trains. So, each person causes less transport-related pollution.

The coefficients of ln_GDP and ln_GDP^2 confirm the nonlinear relationship between income and emissions in both sectors and show an inverted U-shape pattern, where emissions increase with income but decline beyond a certain turning point, consistent with the Environmental Kuznets Curve (EKC).

Energy consumption (ln_energy) is a strong and highly significant driver of emissions in both sectors. Renewable energy use (ln_renew) reduces emissions, confirms the importance of clean energy in mitigating CO_2 emissions.

ln_MAF and ln_SVC**,** where included, are not significant but show expected signs, ln_MAF is negative in TR(6), indicating that manufacturing may reduce transportation emissions.

Adj-R^2 values range from 0.74 to 0.87, indicating that the models explain a large portion of the variation in sector-specific emissions.

3.5.2. Time Series FMOLS Estimation for High-Income Countries

Table 9 presents the results of time series FMOLS regressions using sector-specific CO_2 emissions for high-income countries. Like before we have Models RES(1) to RES(3) and TR(4) to TR(6). As we observe, in both sectors, the coefficients on ln_GDP are negative and those on ln_GDP^2 are positive and statistically significant across all models. The positive sign on the squared income term suggests a U-shaped EKC, contrasts with the traditional EKC's inverted-U form, indicating that after an initial decline, CO_2 emissions rise again at higher income levels. This aligns with findings by Richmond and Kaufmann (2006), who observed similar patterns in high-income countries.

Table 9: Time series FMOLS estimation results for High_Income countries

	RES (1)	RES (2)	RES (3)	TR (4)	TR (5)	TR (6)
const	117.54 **	146.15 ***	115.58 **	152.21 ***	158.16 ***	97.36***
ln_GDP	-30.22 ***	-27.75 ***	-29.72 ***	-30.6 ***	-30.25 ***	19.73***
ln_GDP2	1.44 ***	1.33 ***	1.42 ***	1.42 ***	1.41 ***	0.93***
ln_P	0.33 ***	0.29 ***	0.33 ***	0.11 ***	0.11 ***	0.1***
ln_energy	-0.83 ***	-0.65 ***	-0.83 ***	0.28 ***	0.3 ***	0.16***
ln_renew	-0.12 **	-0.12 **	-0.12 **	0.09 ***	0.09 ***	0.08***
ln_young	0.48	NaN	0.43	1.0 ***	NaN	0.67***
ln_old	1.35 ***	NaN	1.29 ***	-0.22 **	NaN	0.24***
ln_working age	9.38 ***	NaN	9.09 ***	1.33 *	NaN	1.02
lnYouthDep	NaN	-0.96 ***	NaN	NaN	0.71 ***	NaN
lnOldDep	NaN	0.7 ***	NaN	NaN	-0.43 ***	NaN
ln_SVC	NaN	NaN	0.17	NaN	NaN	NaN
ln_MAF	NaN	NaN	NaN	NaN	NaN	0.36***
Adj-R2	0.59	0.55	0.59	0.57	0.56	0.63
Obs	280	280	280	280	280	280

In the residential sector, the elderly population (ln_old) shows strong positive and significant effects on emissions. This indicates that aging contributes to higher residential energy

demand, in contrast, for the transportation sector, older populations (ln_old) have a negative effect, while younger and working-age groups have positive effects. This likely reflects mobility patterns, older individuals in high-income countries are less mobile, while younger and working-age populations are more active users of transportation. This aligns with findings from Kim, Lim, and Jo (2020).

Total population (ln_P) consistently shows a positive and significant effect across all models, confirming that larger populations in high-income countries contribute to increased CO_2 emissions. All models show moderate explanatory power, with adjusted R^2 values ranging from 0.55 to 0.63.

3.5.3. Time Series FMOLS Estimation for Low-Income Countries

To better understand how CO_2 emissions change over time in low-income countries, we run separate time series FMOLS models for residential and transport sectors. This helps us see how factors like income and population affect emissions in each sector.

Table 10: Time series FMOLS estimation results for Low_Income countries

	RES (1)	RES (2)	RES (3)	TR (4)	TR (5)	TR (6)
const	-24.64 ***	-10.93 ***	-24.54 ***	-15.72 ***	-7.32 ***	15.78***
ln_GDP	1.21 ***	2.03 ***	1.21 ***	0.22	0.7 **	0.22
ln_GDP2	-0.08 ***	-0.13 ***	-0.08 ***	0.01	-0.02	0.01
ln_P	0.07 ***	0.07 ***	0.07 ***	-0.05 ***	-0.05***	-0.04***
ln_energy	0.23 ***	0.22 ***	0.23 ***	0.32 ***	0.32 ***	0.32***
ln_renew	-0.36 ***	-0.35 ***	-0.36 ***	-0.09 ***	-0.09 ***	0.09***
ln_young	1.27 ***	NaN	1.28 ***	0.79 ***	NaN	0.79***
ln_old	1.16 ***	NaN	1.16 ***	0.31 ***	NaN	0.31***
ln_working age	2.28 ***	NaN	2.3 ***	1.75 ***	NaN	1.77***
lnYouthDep	NaN	-0.3 **	NaN	NaN	-0.14	NaN
lnOldDep	NaN	0.69 ***	NaN	NaN	0.0	NaN
ln_SVC	NaN	NaN	-0.06	NaN	NaN	NaN
ln_MAF	NaN	NaN	NaN	NaN	NaN	-0.01
Adj-R2	0.72	0.71	0.72	0.79	0.78	0.79
Obs	1193	1193	1193	1193	1193	1193

In residential models (RES 1–3), both GDP (ln_GDP) and GDP squared (ln_GDP2) are significant, with positive GDP and negative GDP^2 coefficients, supporting an inverted U-shape EKC. In the transport models (TR 4–6), the income terms are weaker or insignificant, suggesting no clear EKC pattern for transport emissions in low-income countries. *This is consistent with studies like* Richmond and Kaufmann (2006), *which suggest that low-income countries are often in the rising stage of the EKC.*

All age groups, young (ln_young), old (ln_old) and working-age (ln_working age) have positive and significant effects. These results align with recent findings by Yaqoob et al. (2024), who report that in developing countries, age segments such as 20–34, 35–49, and 50–64 are positively associated with carbon emissions. in our case, even the elderly population contributes positively. This reinforces the idea that in lower-incomecountries, all population groups, including the young and old, are engaged in emission-generating activities.

Additionally, the Adjusted R^2 values, ranging from 0.71 to 0.79 across models, indicate that the explanatory variables collectively explain a large proportion of the variation in CO_2 emissions.

4. Discussion

In this study we analyzed the long-run relationship between CO_2 emissions, economic development, and demographic structure, especially population aging, using panel FMOLS estimation across global, high-income, and low-income countries from 2010 to 2020. By incorporating cubic and quadratic forms of income and age group variables, we investigated the Environmental Kuznets Curve (EKC) and demographic influences on residential and transportation CO2 emissions.

First, our results confirm the presence of nonlinear relationships between income and emissions across all models. In the global analysis, we observe an N-shaped EKC. However, the full shape is not always visible because most countries in the dataset fall into lower or middle-income groups. In low-income countries, emissions keep increasing as income rises, and the turning point occurs at income levels higher than what most of these countries have reached. This suggests that they are still in the early stages of industrialization and emissions growth. Conversely, in high-income countries, we observe an N-shaped EKC, where emissions start to rise again at high income levels.

Second, demographic structure plays an important role. In high-income countries, the elderly population (65 and above) tend to produce less CO_2, especially in the transport sector, likely due to reduced mobility and lower consumption patterns. This supports earlier research by Kim, Lim, and Jo (2020). In contrast, young population, is positively associated with emissions, especially in transportation, likely due to higher travel activity and consumption.

However, in low-income countries, all age groups, including the elderly, show a positive and significant relationship with emissions in both sectors. This result aligns with Yaqoob et al. (2024), who found that even older age groups in developing countries contribute to emissions due to weaker infrastructure, limited access to clean technologies, and less efficient energy use.

5. Conclusion

Our results are important for shaping future policies. The inverted U-shaped EKC in global data and the N-shaped EKC in high-income countries show that economic growth alone is not enough to reduce emissions. Countries need to support growth with cleaner energy and new technology to cut emissions as income increases.

Demographic changes, especially aging, impact emissions in different ways across countries. In high-income countries, older populations usually emit less due to lower mobility and consumption. But in developing countries, aging doesn't always reduce emissions, since older adults often remain active and live in areas with less efficient infrastructure.

In low-income countries, the ongoing rise in emissions alongside income and across all age groups shows the importance of investing in clean energy and sustainable infrastructure early in their development path.

Overall, this study shows that both economic growth and population age need to be considered when making environmental policies. Reducing income gaps, improving access to clean energy, and planning for the needs of different age groups can help countries grow in a more sustainable way.

References

1. **Apeti, A. E.**, Bambe, B. W. W., Edoh, E. D., & Ly, A. (2025). Wealth inequality and carbon inequality. *Ecological Economics, 227*, 107633. https://doi.org/10.1016/j.ecolecon.2022.107633

2. **de Bruyn, S. M.**, van den Bergh, J. C. J. M., & Opschoor, J. B. (1998). Economic growth and emissions: Reconsidering the empirical basis of environmental Kuznets curves. *Ecological Economics, 25*(2), 161–175. https://doi.org/10.1016/S0921-8009(97)00178-X

3. **Fakher, H. A.**, Ahmed, Z., Acheampong, A. O., & Nathaniel, S. P. (2023). Renewable energy, nonrenewable energy, and environmental quality nexus: An investigation of the N-shaped Environmental Kuznets Curve based on six environmental indicators. *Energy, 263*, 125660. https://doi.org/10.1016/j.energy.2022.125660

4. **Filonchyk, M.**, Peterson, M. P., Zhang, L., Hurynovich, V., & He, Y. (2024). Greenhouse gases emissions and global climate change: Examining the influence of CO_2, CH_4, and N_2O. *Science of the Total Environment, 905*, 173359. https://doi.org/10.1016/j.scitotenv.2024.173359

5. **Grossman, G. M.**, & Krueger, A. B. (1995). Economic growth and the environment. *The Quarterly Journal of Economics, 110*(2), 353–377. https://doi.org/10.2307/2118443

6. **Hamza, N.**, & Gilroy, R. (2011). The challenge to UK energy policy: An ageing population perspective on energy saving measures and consumption. *Energy Policy, 39*(2), 782–789. https://doi.org/10.1016/j.enpol.2010.10.052

7. **Jahanger, A.**, Usman, B., & Balsalobre-Lorente, D. (2023). The role of environmental technologies and clean energy transition in the N-shaped environmental Kuznets curve: Evidence from the top ten carbon-emitting countries. *Environmental Technology & Innovation, 32*, 103463. https://doi.org/10.1016/j.eti.2023.103463

8. **Jin, Y.**, Sharifi, A., Li, Z., Chen, S., Zeng, S., & Zhao, S. (2024). Carbon emission prediction models: A review. *Science of the Total Environment, 903*, 172319. https://doi.org/10.1016/j.scitotenv.2024.172319

9. **Kao, C.** (1999). Spurious regression and residual-based tests for cointegration in panel data. *Journal of Econometrics, 90*(1), 1–44. https://doi.org/10.1016/S0304-4076(98)00023-2

10. **Kim, J.**, Lim, H., & Jo, H.-H. (2020). Do aging and low fertility reduce carbon emissions in Korea? Evidence from IPAT augmented EKC analysis. *International Journal of Environmental Research and Public Health, 17*(9), 2972. https://doi.org/10.3390/ijerph17082972

11. **Levin, A.**, Lin, C.-F., & Chu, C.-S. J. (2002). Unit root tests in panel data: Asymptotic and finite-sample properties. *Journal of Econometrics, 108*(1), 1–24. https://doi.org/10.1016/S0304-4076(01)00098-7

12. **Liddle, B.**, & Lung, S. (2010). Age-structure, urbanization, and climate change in developed countries: Revisiting STIRPAT for disaggregated population and consumption-related environmental impacts. *Population and Environment, 31*(5), 317–343. https://doi.org/10.1007/s11111-010-0101-5

13. **Long, Y.**, Feng, J., Sun, A., Wang, R., & Wang, Y. (2023). Structural Characteristics of the Household Carbon Footprint in an Aging Society. *Sustainability, 15*(17), 12825. https://doi.org/10.3390/su151712825

14. **Maddala, G. S.**, & Wu, S. (1999). A comparative study of unit root tests with panel data and a new simple test. *Oxford Bulletin of Economics and Statistics, 61*(S1), 631–652. https://doi.org/10.1111/1468-0084.0610s1631

15. **Marselina, M.**, & Prasetyo, T. J. (2023). Does the "Environmental Kuznets Curve" Phenomenon Happen in High, Medium, and Low Income Countries? *International Journal of Energy Economics and Policy, 13*(1), 544–551. http://hdl.handle.net/11159/59384

16. **Mbiankeu Nguea, S.** (2023). The role of working-age population in environmental sustainability: Evidence from panel data analysis. *Nature Sustainability*, Article 100005. https://doi.org/10.1016/j.nxsust.2023.100005

17. **Mitić, P.**, Fedajev, A., Radulescu, M., & Rehman, A. (2023). The relationship between CO_2 emissions, economic growth, available energy, and employment in SEE countries. *Environmental Science and Pollution Research, 30*(6), 16140–16155. https://doi.org/10.1007/s11356-022-23356-3

18. **Pedroni, P.** (2000). Fully modified OLS for heterogeneous cointegrated panels. *Advances in Econometrics, 15*, 93–130. https://doi.org/10.1016/S0731-9053(00)15004-2

19. **Pedroni, P.** (2001). Purchasing power parity tests in cointegrated panels. *The Review of Economics and Statistics, 83*(4), 727–731. https://doi.org/10.1162/003465301753237803

21. **Pesaran, M. H.** (2007). A simple panel unit root test in the presence of cross-section dependence. *Journal of Applied Econometrics, 22*(2), 265–312. https://doi.org/10.1002/jae.951

22. **Planet App.** (2020). *Carbon Footprint and Age: How Old Are the Most Polluting People?* https://theplanetapp.com/carbon-footprint-and-age/?lang=en

23. **Raupach, M. R.**, Marland, G., Ciais, P., Le Quéré, C., Canadell, J. G., Klepper, G., & Field, C. B. (2007). Global and regional drivers of accelerating CO_2 emissions. *Proceedings of the National Academy of Sciences, 104*(24), 10288–10293. https://doi.org/10.1073/pnas.0700609104

24. **Richmond, A. K.**, & Kaufmann, R. K. (2006). Is there a turning point in the relationship between income and energy use and/or carbon emissions? *Ecological Economics, 56*(2), 176–189. https://doi.org/10.1016/j.ecolecon.2005.01.011

25. **Stern, N.** (2007). *The Economics of Climate Change: The Stern Review*. Cambridge University Press. https://doi.org/10.1017/CBO9780511817434

26. **UNDP.** (2022). *How large are inequalities in global carbon emissions, and what to do about it?* United Nations Development Programme. https://hdr.undp.org/content/how-large-are-inequalities-global-carbon-emissions-and-what-do-about-it

27. **Zou, J.**, Zhao, X., & Zhang, M. (2022). Environmental Kuznets Curve in the context of sustainable development: A comprehensive review. *Journal of Cleaner Production, 341*, 130912. https://doi.org/10.1016/j.ecolmodel.2020.109289